

Preterm Birth Prediction Using GraphSAGE and Electrohysterogram Data

Nandini Kuppala

*Amrita School of Artificial Intelligence,
Coimbatore,
Amrita Vishwa Vidyapeetham, India*
ORCID: 0009-0008-8746-1227

Monish GV

*Amrita School of Artificial Intelligence,
Coimbatore,
Amrita Vishwa Vidyapeetham, India*
ORCID: 0009-0000-5586-7558

Nivedha V

*Amrita School of Artificial Intelligence,
Coimbatore,
Amrita Vishwa Vidyapeetham, India*
ORCID: 0009-0000-6999-2149

Balu Pinnenti

*Amrita School of Artificial Intelligence,
Coimbatore,
Amrita Vishwa Vidyapeetham, India*
ORCID: 0009-0008-2828-1226

Amrutha V

*Amrita School of Artificial Intelligence,
Coimbatore,
Amrita Vishwa Vidyapeetham, India*
ORCID: 0000-0002-8969-2810

Abstract—Preterm birth (PTB) affects around 15 million infants every year and causes major health issues, making it a worldwide health concern [4]. The demand for precise early prediction methods has motivated researchers to explore the use of Electrohysterogram (EHG) signals. EHG is a promising approach for predicting preterm labor as it is a non-invasive and affordable method for monitoring uterine activity. This study aims to build a strong PTB prediction model with explainable AI using the publicly accessible TPEHG dataset (300 recordings, 38 preterm). The validation of the model is performed with the TPEHGT dataset (13 preterm and 13 term samples). We employed a graph neural network (GNN) that efficiently aggregates local neighborhood information in graphs and is based on the GraphSAGE architecture. The GraphSAGE model achieved 99.2% accuracy by training with stratified 10-fold cross-validation. This was validated through confusion matrices, ROC, and precision-recall curves. In order to guarantee model transparency, the most important factors influencing predictions were found using Shapley Additive explanations (SHAP) analysis, which offered insightful information about the decision-making process. This study emphasizes the value of explainable AI in enhancing clinical decision-making, as well as the possibility of GNNs for PTB prediction.

Index Terms—EHG, Preterm, GraphSAGE, ADASYN, SHAP, TPEHG

I. INTRODUCTION

According to the World Health Organization (WHO), Premature birth is defined as the delivery of a baby before the mother has reached 37 weeks of pregnancy. Prematurity accounts for the majority of neonatal intensive care unit (NICU) admissions [6]. Worldwide, 11.1% of fetuses are born before their due date. Families and healthcare systems around the world are concerned about this issue because of the financial, emotional, and physical difficulties. Early detection of preterm labor is crucial for implementing timely interventions that can improve maternal and neonatal outcomes. The prevalence of premature delivery can be decreased

with the early detection and appropriate treatment of various conditions, including pregnancy-induced hypertension, urinary tract infections, anemia, and systemic sickness [5]. Traditional diagnostic techniques, such as measuring cervical length and analyzing maternal history, have not been very successful in predicting preterm births at an early stage.

EHG signals, have become a useful and potential method for predicting preterm birth because they use surface electrodes to record the electrical activity of the uterine muscles [8]. The EHG method establishes a link between uterine contraction and electrical activity, providing a novel way to improve preterm diagnosis.

Recent research efforts have shown the potential of machine learning and deep learning models in EHG analysis signals to classify term and preterm deliveries. The recent studies uses both traditional machine learning approaches and deep learning approaches including Convolutional Neural Network (CNN) and Spiking Neural Network (SNN). We present a model that applies GNNs to term and preterm classification. In contrast to traditional models, GNNs are especially well-suited for capturing complicated relationships within the data by virtue of graph-based structures which allow the model to capture patterns in the EHG signal. Previous research has focused on various feature extraction techniques and classifier combinations, often achieving promising results using ensemble models and deep learning methods. However, most of these approaches are not transparent about their decision-making processes, which is crucial in clinical applications.

We analyze the results of our model through explainable AI methods like SHAP. This explanation plays an important role in the real-world application of the model and strengthens the connection between the physicians. Our method seeks to integrate the strength of Graph Neural Networks with the interpretability of SHAP to generate a strong, interpretable

PTB prediction model.

II. LITERATURE REVIEW

Several investigations have investigated the effective forecasting methodologies of preterm labor based on EHG signals. The empirical mode decomposition was one of the approaches for extracting linear and nonlinear features of uterine electrical activity for a remarkably high accuracy of 99.23% with an Artificial Sampling technique-enhanced Random Forest classifier [1]. Another study used the CNN-based MobileNet-V2 model for feature extraction and compared several classifiers, showing that the Quadratic Discriminant classifier was the best with 99.8% accuracy [2]. Recurrent neural networks combined with short-time Fourier transform and clinical data demonstrated that spectral patterns in EHG signals were more effective for prediction than temporal patterns [7].

A hybrid machine learning approach, Enhanced Sheep Flock Optimized Hybrid Extreme Artificial Neural Learning Network, was found to outperform traditional approaches such as SVM and Random Forests with better accuracy, specificity, recall and F-measure [10]. In the same way, a deep learning approach that combined SNN with the Seagull Optimization Algorithm was able to outperform traditional approaches with higher precision, better recall, and reduced processing time [11].

Other studies tried to establish associations between gestational age and EHG time domain characteristics. Based on this line of research, m-order, Zero Crossing, MAV and difference absolute standard deviation value have been found to have considerable linear correlations to recording [13]. Overall these studies collectively affirm the effectiveness with which advanced signal analysis and learning machine techniques for predicting preterm labor.

III. METHODOLOGY

A. Dataset

TPEHG DB is the term-preterm electro-hysterogram database designed for uterine EHG recordings to study uterine electrical activity in pregnancy. Developed at the University Medical Centre Ljubljana between 1997 and 2005, it is a repository supporting research into predicting preterm delivery. The dataset includes recordings from 300 pregnant women, including both term and preterm deliveries.

1) *Overview of the Dataset:* The key characteristics of the TPEHG DB are as follows:

- Total Records: 300 uterine EMG recordings.
- Term Deliveries: 262 records (gestation period > 37 weeks).
- Preterm Deliveries: 38 records (gestation period ≤ 37 weeks).
- Gestation Stages:
 - Before the 26th week: 162 records.
 - During or after the 26th week: 138 records.
- Recording Duration: Each recording lasts 30 minutes.
- Sampling Rate: Signals were sampled at 20 Hz per channel with 16-bit resolution.

2) *Recording Configuration:* The uterine EMG signals were captured using four electrodes placed symmetrically around the navel. The potential differences between these electrodes were used to create three recording channels. where Channel 1 (s1) represents the difference between E1 and E2, Channel 2 (S2), Channel 2 (S2), represents the difference between E2 and E3 and Channel 3 (S3), indicating the difference between E4 and E3.

3) *Clinical Information:* The dataset is supplemented with clinical metadata for each record, which includes:

- Maternal details: age, gestation duration, and pregnancy duration.
- Clinical observations: hypertension, diabetes, smoking status, and placental position.
- Delivery specifics: type of delivery and prior obstetric history (e.g., parity, abortions).

4) *Data Summary Table:* Table I provides a detailed summary of the dataset distribution based on gestation stages and delivery outcomes.

TABLE I
SUMMARY OF DATASET DISTRIBUTION

Group	Before 26 Weeks	After 26 Weeks
Term Deliveries (>37 weeks)	143	119
Preterm Deliveries (≤ 37 weeks)	19	19

B. Preprocessing

EHG signals are prone to noise and have a low amplitude. Preprocessing is required to guarantee the dependability and quality of the EHG signals for analysis. In this study first, we normalized the raw signal to a range of [0,1] to reduce amplitude variability caused by physiological differences. The frequency range of uterine contractions is then separated from noise, including high-frequency interference and maternal and fetal ECG components, using bandpass filter (0.3–3.0 Hz). Then we used wavelet denoising using discrete wavelet transforms like db4 which reduces transient noise while preserving important signal features through soft thresholding [17]. Then we identified and removed outliers using z-score filtering, which excludes data points deviating significantly from the mean, mitigating the impact of anomalies caused by electrode artifacts or sudden movements. We applied median filtering to remove impulsive noise and moving average smoothing to emphasize trends and reduce high-frequency fluctuations. Finally, we applied standardization which centers the data around zero mean and scales it to unit variance. These preprocessing steps improve signal quality, reduce noise, and preserve critical uterine activity patterns, making the EHG signals suitable for further analysis and interpretation. The plot of a signal before and after preprocessing is shown in Fig. 1.

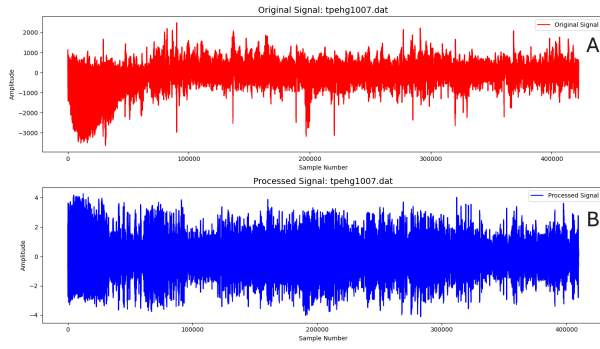


Fig. 1. (A) EHG signal before preprocessing, (B) EHG signal after preprocessing

C. Feature Extraction

Feature Extraction plays a crucial step in the study of EHG signals to find important patterns that characterize uterine activity. We extracted frequency domain features, time domain features, nonlinear features, and time-frequency features that are useful for distinguishing between term and preterm births [18].

Statistical features like mean, variance, dispersion, and kurtosis describe the central tendency, asymmetry, peakedness, and skewness of the signal distribution, respectively. The average signal amplitude is reflected in the mean that gives the intensity of uterine contractions on average, the variance calculates the variability in the signal which is a way of quantitatively describing how much the signal fluctuates which is important when determining the stability of uterine activity during labor. The skewness and kurtosis are used in describing the signal distribution shape which can give some insight into anomalies in contraction patterns.

Another key statistical measurement of are Interquartile Range (IQR) and Root Mean Square (RMS). RMS measures the strength of the contraction by showing the total amount of energy by the signal. IQR takes into account the middle 50% of data, thereby cutting down the impacts of outliers in the signal with information about how variable the signal is. It can also indicate the presence or absence of pathological uterine activity. The zero crossings as well as the envelope of the signal indicate its oscillatory behavior and variations in amplitude that change in time. Zero-crossings represent a count of how many times the signal crosses through the zero line of amplitude to indicate the frequency of contractions. An envelope smooth represents the changes of amplitude that signify the signal that is needed in understanding temporal contractions. PSD in defined bandwidths: (0.3–0.6 Hz), (0.6–1.0 Hz). Dominant frequency, mean, and total power within spectral entropy are crucial for quantifying the energy distribution within contraction-related frequency ranges. PSD is a measure of signal power distributed across different frequency bands. We analyzed PSD, particularly in the ranges 0.3–0.6 Hz and 0.6–1.0 Hz, which are of high significance for understanding uterine contractions. Dominant frequency and mean frequency are both related to the prevailing oscillations

in the signal and total power measures the total energy present in the signal. Spectral entropy measures the complexity of the signal, highlighting information regarding how unpredictable uterine Contractions can be.

Wavelet-based features, including wavelet entropy and wavelet energy at different levels, from Level 1 to Level 5, capture the energy of the signal at different scales while retaining the temporal and frequency information needed to analyze non-stationary EHG signals. Wavelet energy can reveal transient events and changes in uterine activity that are not possible with traditional Fourier methods, making these features especially useful in understanding dynamic changes in uterine activity. Dynamic behavior of uterine activity The features presented are instantaneous amplitude and frequency measurement, such as instantaneous amplitude mean, instantaneous amplitude standard deviation, instantaneous frequency mean, and instantaneous frequency standard deviation. These help reveal how the changes in the amplitudes and frequencies of contractions during pregnancy could serve as indicators for uterine health and responsiveness.

Finally, we extracted recurrence quantification analysis (RQA) features such as recurrence rate, determinism, laminarity, and L max. Recurrence rate and determinism tell about the repeated and organized pattern, while laminarity and L max show long-term behavior and stability. These metrics inform about patterns and predictability within the EHG signals that may be of importance in identifying abnormalities in uterine contractions. Collectively, these extracted features provide a comprehensive understanding of EHG signals by capturing both the statistical and dynamic uterine activity patterns. To enhance the analysis, we combine these signal-based features with some clinical features taken from the linked .hea files in the dataset. These clinical features include key maternal and pregnancy-related factors such as gestation (duration of pregnancy), parity (number of previous births), recording time, age, weight, presence of hypertension, history of abortions, diabetes, placental position, and instances of bleeding during the first and second trimesters. Additionally, factors like cervical funneling and smoking habits were also considered. We used label encoding to transform categorical data into a numerical format so that these clinical features could be included in the study. The missing values in the dataset were addressed by imputation, where the mode was applied to categorical features and the median to numerical features. By doing this, possible biases brought on by incomplete records are reduced and data consistency is guaranteed [16].

D. Feature Selection

Feature selection is an important step to enhance model performance and reduce redundancy. In this study, we combined machine learning and statistical methods to construct a thorough feature selection approach. To make sure the remaining characteristics offered distinct and independent information, we first employed correlation-based filtering to find and exclude strongly associated features with a threshold of 0.9. Then we employed an ensemble approach to feature

selection by combining the results from three different models: Random Forest, Recursive Feature Elimination (RFE), and L1 Regularization (Lasso). Every model offers a distinct viewpoint on feature significance, enabling us to obtain a thorough understanding of the dataset’s essential characteristics.

RFE iteratively eliminates the least significant features according to their contribution to model performance. It ranks the features and keeps those that have a major influence on prediction accuracy using a logistic regression model. Lasso penalizes the feature coefficients, it essentially reduces the weights of features that aren’t significant to zero. In order to rank features according to their capacity to lower impurity in decision trees, we used random forest feature importance. By capturing non-linear correlations and interactions between characteristics, this tree-based approach offers a greater understanding of their significance for predicting preterm birth. In order to ensure a consistent baseline for comparison, we first used Min-Max scaling to scale the significance scores from each model to a common range. In order to capitalize on the advantages of each approach, the scaled scores were then combined (Fig. 2) This combined score provided a more robust assessment of feature importance, as it integrated the insights gained from different modeling techniques.

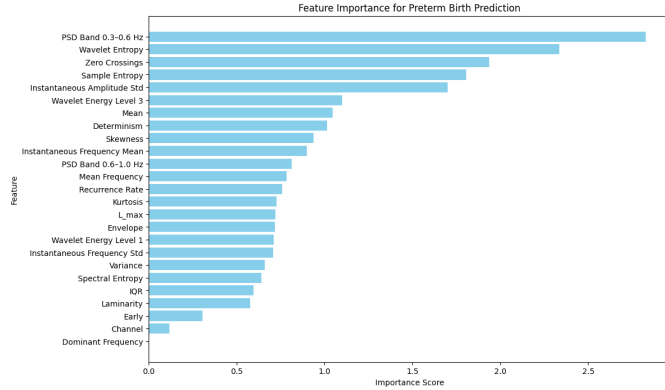


Fig. 2. This bar chart illustrates the importance scores of various features extracted from the dataset. The features are ranked in descending order of importance, with the most influential features at the top.

E. Data Balancing

In this study, we used Adaptive Synthetic Sampling (ADASYN) to handle class imbalance in our dataset [14]. Class imbalance occurs when instances of the minority class (preterm) are substantially lower than those of the majority class (term). This can lead to biased models that favor the majority class and result in poor performance of the minority class. We used ADASYN on training data after splitting the train and test data (80%, 20% respectively) to avoid bias influenced by synthetic samples on the test data. ADASYN is useful because it uses the local distribution of minority instances to create synthetic samples of the minority class. By using an adaptive strategy, the model is better able to comprehend the features of premature births by concentrating on producing more examples in which the minority class is

underrepresented. Focusing on regions of the feature space that present classification issues is one of ADASYN’s main advantages over other oversampling techniques like the Synthetic Minority Over-sampling Technique (SMOTE). That’s why ADASYN was preferred over SMOTE in many studies [3], [9], [15]. Moreover, ADASYN helps reduce the risk of overfitting, a common problem with SMOTE that arises when an excessive number of synthetic samples are generated in densely populated portions of the feature space.

F. Model Architecture and Mechanism: GraphSAGE

Overview of GraphSAGE

GraphSAGE is a scalable, semi-supervised graph neural network. It combines and aggregates feature information from a node’s local neighborhood to learn node representations. Traditional GNNs operate on the entire graph, whereas GraphSAGE samples a fixed-size subset of neighbors, which is computationally efficient and makes it ideal for large graphs. Instead of learning individual embeddings, it learns aggregation functions. This allows it to generalize successfully to unseen nodes. GraphSage was initially proposed in the study [19]. GraphSage was used in [20] for Recognition of Autism Spectrum Disorder.

GraphSAGE operates in three key steps: *sampling*, *aggregation*, and *update*. During the sampling step, a fixed-size subset of neighbors is selected for each node. Next, the aggregation step combines features of the sampled neighbors using predefined functions like mean, LSTM-based aggregation, or pooling. In this work, we utilize SAGEConv with mean aggregation function, which efficiently performs this aggregation. Finally, during the update step, the node’s feature representation is updated by combining its original features with the aggregated neighborhood features. This process is repeated across multiple layers to capture higher-order neighborhood information.

Mathematically, the operation for each layer k is expressed as:

$$h_v^{(k)} = \sigma \left(W^{(k)} \cdot \text{AGGREGATE}^{(k-1)} \left(h_v^{(k-1)}, \{h_u^{(k-1)} : u \in \mathcal{N}(v)\} \right) \right) \quad (1)$$

Where:

- $h_v^{(k)}$ represents the feature vector of node v at the k -th layer,
- $\mathcal{N}(v)$ signifies the neighboring nodes of v ,
- AGGREGATE combines the features of neighbors $h_u^{(k-1)}$ with the features of node $h_v^{(k-1)}$,
- $W^{(k)}$ denotes the trainable weight matrix at layer k , and
- σ refers to a non-linear activation function, such as ReLU.

Model Architecture

The model architecture operates as follows: First, Graphs are constructed for training and validation sets using cosine similarity between feature vectors. Each patient is represented by a node, characterized by a 38-dimensional feature vector, while edges are formed on the basis of cosine similarity (greater than 0.85) among the feature vectors of patients and make sure patients with similar features are connected. The

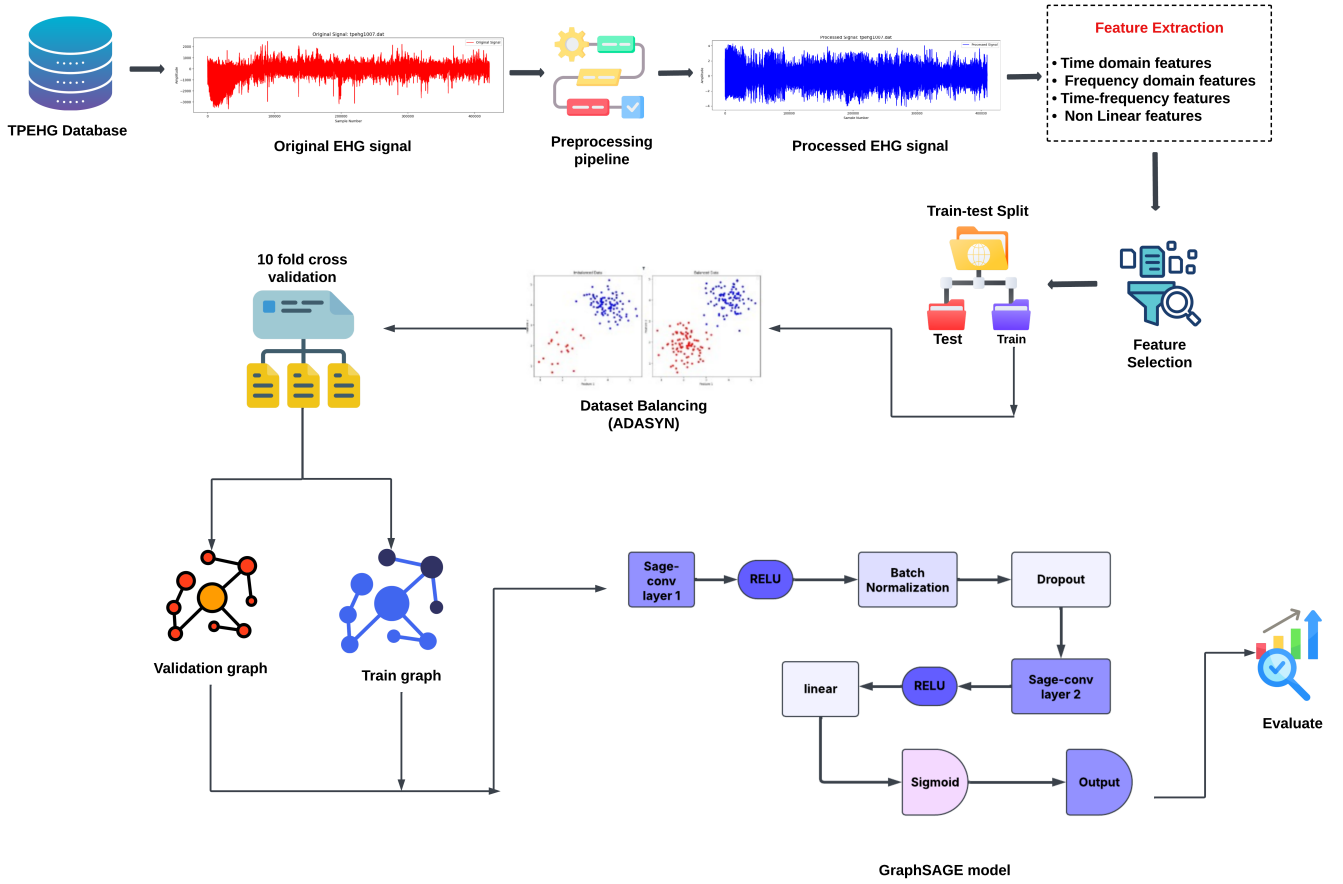


Fig. 3. Methodology Overview

model begins with an input layer, which can accept standardized feature vectors as its input. Then the model contains two SAGEConv layers such that the gathered information from sampling the neighbors for each node using mean pooling followed by applying a linear transformation followed by an application of the ReLU function. The SAGEConv layers are followed by the Batch Normalization and Dropout layers to enhance the stability and generalization of the model during training. Finally, a fully connected output layer produces the logits for binary classification. The trained model enables real-time inference by identifying similar historical patients, aggregating their clinical features, and predicting the probability of preterm birth. This approach ensures scalability, better generalization, and context-aware predictions.

Training Methodology

A stratified K-Fold cross-validation method is used to train the model. The model is trained for 100 epochs for every fold. We used the Adam optimizer with a weight decay of 5×10^{-4} and a learning rate of 0.01. After each forward pass, gradients were computed, and the model parameters were subsequently updated. To track performance across epochs, the training and validation accuracies are computed.

All the steps discussed so far are summarized in the block diagram Fig. 3, clearly outlining the processes of preprocessing, feature extraction, feature selection, data balancing, model training, and evaluation.

G. Validation Using an External Dataset

To increase the number of preterm samples and test our model on unseen data, we merged the TPEHG dataset with the Term-Preterm ElectroHysteroGram Dataset with Tocogram (TPEHGT). The TPEHGT dataset contains 26 uterine EHG recordings, each 30 minutes long and with four signals. It contains 13 preterm records from eight pregnancies and 13 term records from ten pregnancies that ended in spontaneous term delivery. The recording protocol of TPEHGT, such as electrode placement, was the same as that used for the TPEHG database which provides compatibility between the datasets.

But some clinical features present in TPEHG were not present in TPEHGT, such as cervical funneling, bleeding in the first and second trimesters, hypertension, and diabetes. The feature vector size was therefore decreased from 38 to 33 dimensions to provide compatibility between the datasets. For the TPEHGT dataset, we followed the same preprocessing, feature extraction, and feature selection procedures that have

been previously explained. After merging the two datasets, we repeated the training process using 10-fold cross-validation. The results of this analysis will be discussed in the next section.

IV. RESULTS

A. Results of the model Trained on TPEHG dataset

Accuracy

Accuracy is the ratio of the number of correct predictions of instances to the total number of predictions. The accuracy of the model across 10 folds is shown in Fig. 4 .

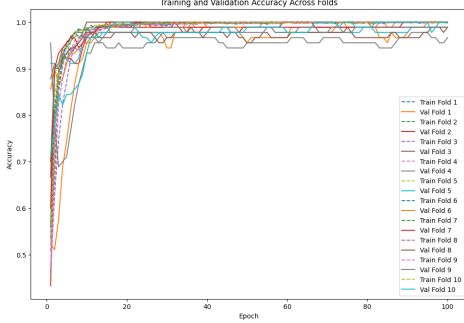


Fig. 4. Accuracy across 10-fold cross-validation

Confusion Matrix

The confusion matrix (Fig. 5) evaluates the model's classification performance by comparing actual and predicted outcomes. The model correctly classified 785 term cases and 108 preterm cases, with only 1 false positive and 6 false negatives.

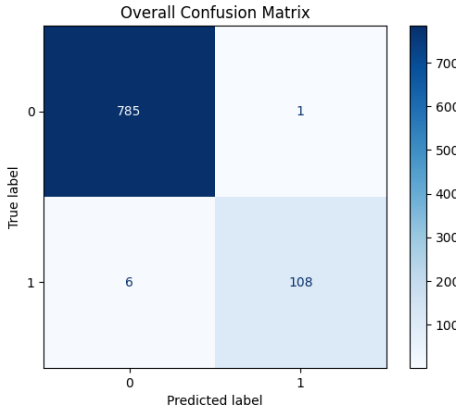


Fig. 5. Confusion matrix

Training Loss

Training loss refers to the error measured when the model makes predictions on the training data. As the model undergoes each forward and backward pass, the loss is updated. The loss gradually decreases as the model learns to predict more accurately (Fig. 6).

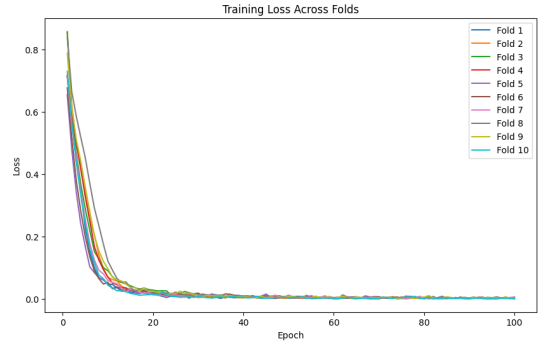


Fig. 6. Training loss across 10-fold cross-validation

ROC curve

The ROC curve (Fig. 7) demonstrates the capacity of the model to classify term and preterm deliveries, with an AUC of 1.00, reflecting perfect classification. The steep rise towards (0,1) shows high true positive rate with low false positives, reflecting good discriminative power. Although this performance is desirable, the near-perfect AUC can suggest possible overfitting, and further validation on an external data set is necessary to confirm generalizability.

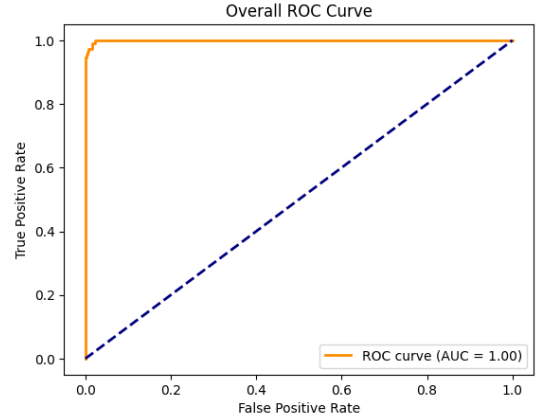


Fig. 7. ROC curve

Precision-Recall Curve

Precision-Recall curve (Fig. 8) demonstrates good performance of the model in detecting preterm births with high precision and recall at all thresholds. The curve stays close to (1.0,1.0), implying few false positives and effective handling of class imbalance. A gentle decrease at very high recall values implies a minor trade-off between recall and precision. Overall, the model shows consistent classification.

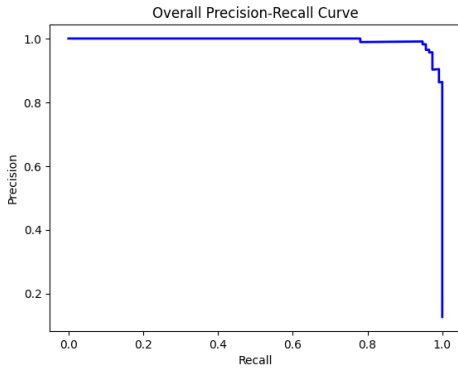


Fig. 8. Precision-Recall curve

Classification Report

The classification report summarizes key metrics used to evaluate the performance of a classification model, including recall, precision, F1-score, and accuracy for each class. It also includes overall averages, such as macro and weighted averages. We present the classification report as a heatmap for better visual interpretation (Fig 9). Each cell represents the value of a specific metric for a given class or average, with color gradients indicating higher or lower values. In the case of term births (Class 0), precision is 0.99, recall is 1.00, and the F1-score is 1.00, indicating that the model correctly identifies almost all full-term births with minimal false positives. In the case of preterm births (Class 1), precision is 0.99, recall is 0.95, and the F1-score is 0.97, suggesting that while the model effectively detects most preterm cases, a slightly lower recall indicates a few false negatives. The macro-averaged recall of 0.97 and the weighted F1-score of 0.99 indicate strong overall classification performance, with well-balanced predictions across both classes.

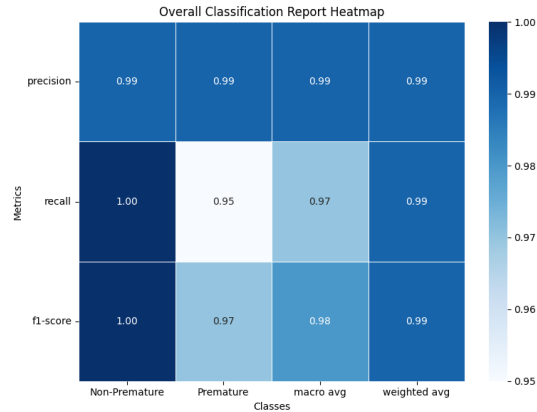


Fig. 9. Classification report heatmap

Overall Results

The model had a total accuracy of 99.22%, proving to be reliable. The sensitivity (recall) was 94.74%, showing good capability to identify preterm cases, while specificity was 99.87%, which prevented low false positive predictions.

B. SHAP

SHAP is an interpretability tool that assesses the impact of each feature on the model's output and assigns importance scores. It identifies the most influential features affecting preterm birth predictions, making it valuable for healthcare applications. SHAP ensures fairness by detecting biases and promoting transparency in predictions. By visualizing SHAP values, we can refine the dataset, validate predictions, and enhance trust in the model. The SHAP results are presented in Fig. 10 and Fig. 11.

SHAP Summary Plot : The SHAP summary plot, shown in Fig. 10, highlights the features that most influence the model's predictions. The x-axis represents SHAP values, and the y-axis lists all features. Features are ranked by their SHAP values, with "Gestation" being the most influential. Positive SHAP values increase the model's output, while negative values reduce it. The color gradient, ranging from red to blue, indicates higher to lower values. For example, high values of "Gestation", "Funneling" and "Bleeding in second trimester" strongly increase the prediction, while low values reduce it. "Wavelet Entropy" and "Weight" have mixed effects depending on their values. This plot visually explains the role of each feature and its impact on predictions.

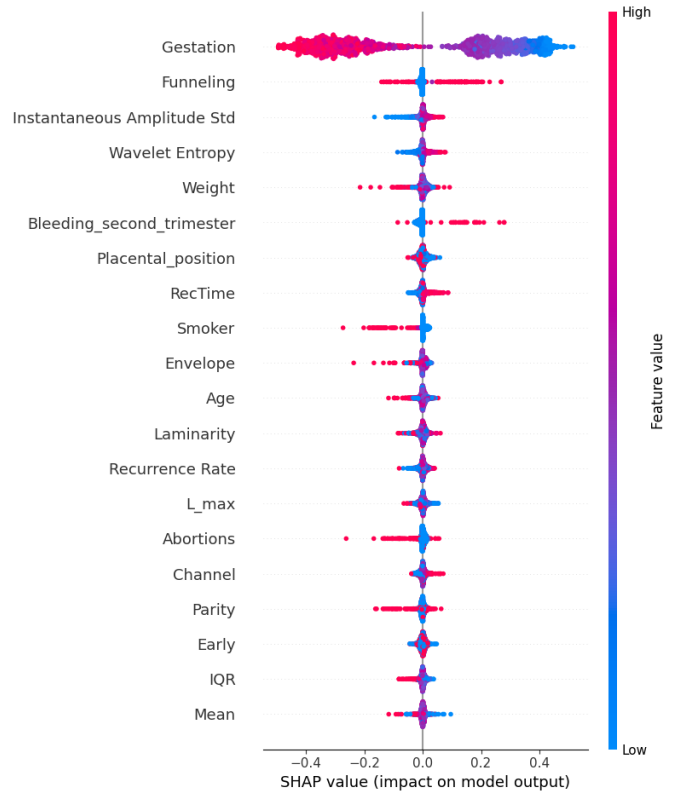


Fig. 10. SHAP summary plot

SHAP Waterfall Plot: The SHAP waterfall plot (Fig. 11) shows how individual features contribute to a model's prediction for a specific instance. It breaks down the prediction step

by step, showing the cumulative effect of each feature. Red bars indicate positive contributions, while blue bars indicate negative contributions. In this plot, each number next to a feature shows how much that feature adds to the final prediction. For example, "+0.32" next to "Gestation" means this feature adds 0.32 to the model's prediction. The starting point is the model's initial expected value (0.51), and as you move through the plot, the contributions of each feature are added or subtracted. The final value, $f(x) = 1$, represents the model's predicted outcome after considering all the features.

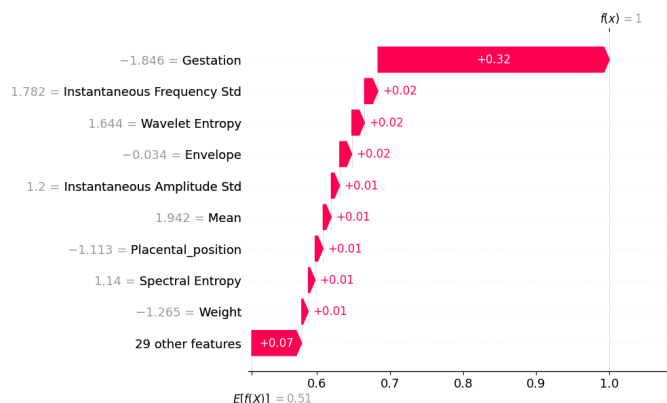


Fig. 11. SHAP waterfall plot

C. Results of the Model Trained on the Merged TPEHG and TPEHGT Datasets

Accuracy

The average accuracy of the model across 10 folds (Fig. 12) is 97.65 which is less than the model trained only on TPEHG dataset.

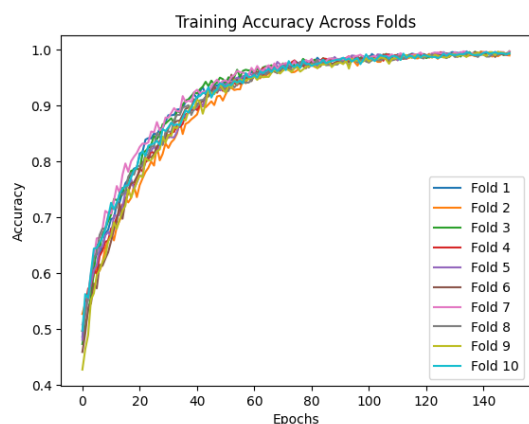


Fig. 12. Accuracy across 10-fold cross-validation

Confusion Matrix

The confusion matrix (Fig. 13) shows good classification performance with 805 true negatives and 150 true positives. The model inaccurately classified 20 term births as preterm (false positives) and only failed to detect 3 preterm cases (false negatives), maintaining good recall for preterm detection.

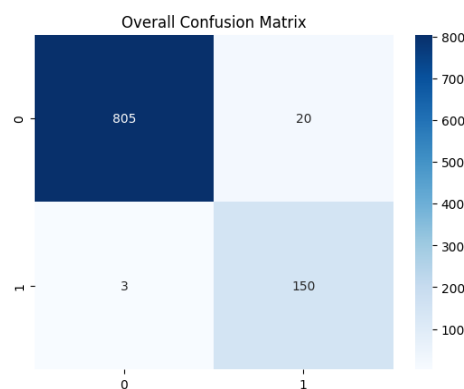


Fig. 13. Confusion matrix

Classification Report

Classification report heatmap of the model trained on merged data (TPEHG + TPEHGT) is shown in Fig. 14. In the case of term births (Class 0), precision was 1.00, recall was 0.97, and F1-score was 0.98. In the case of preterm births (Class 1), precision was 0.85, recall was 0.98, and the F1-score was 0.91. High macro-averaged precision of 0.92 and weighted F1-score of 0.97 also suggest good overall predictive performance. The slightly lower precision for preterm indicates some false positives but high recall indicates few false negatives.

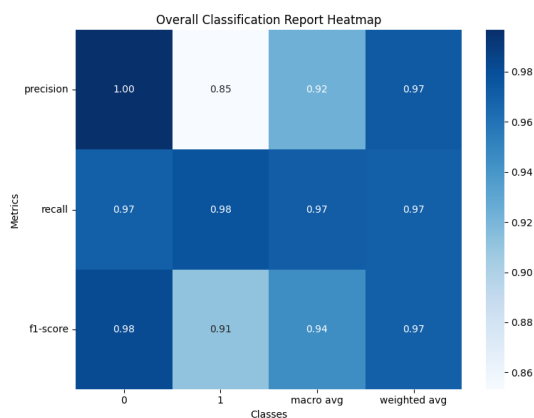


Fig. 14. Classification report heatmap

Overall Results

The model achieved a total accuracy of 97.65%, which is reliable. The sensitivity (recall) was 98.04%, which performed better than the model trained only on TPEHG data, reflecting good capability in detecting preterm cases. Specificity was 97.58%, which was slightly lower compared to the TPEHG-only model.

The reduced accuracy of the model trained on merged TPEHG and TPEHGT data, compared to the TPEHG-only model, may be due to the omission of important clinical features such as Funneling and Bleeding during the second trimester, which have a major impact on predictions. The

SHAP summary plot (Fig. 10) indicates that these features significantly contribute to preterm birth predictions. Since, these features were not present in the TPEHGT dataset, they were also dropped from the TPEHG dataset to ensure consistency.

V. CONCLUSION

This study introduces the use of the GraphSAGE model for predicting preterm birth based on EHG signals, achieving an impressive 99% accuracy with stratified 10-fold cross-validation. The model's ability to aggregate local neighborhood information through GraphSAGE enhances accuracy and robustness compared to previous methods. A comprehensive feature selection approach combining Random Forest, RFE, and L1 Regularization was used to select the most informative and independent features. Additionally, Explainable AI (SHAP) was integrated to improve model interpretability, providing insights into feature importance and the impact of each feature on predictions. For generalization, we evaluated the model using a combined dataset (TPEHG + TPEHGT), which resulted in a slightly lower accuracy (97.65%) than the model trained solely on TPEHG but achieved a higher sensitivity (98.04%). Future research will focus on testing the model with different datasets and exploring deeper graph-based methods to further improve interpretability and clinical reliability.

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